

Discrete Mathematics

Solved Problem Set

CSE230 — Discrete Mathematics
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Section 1: Logic and Propositional Calculus

Problem 1 — Truth Table and Classification

Construct the truth table for the proposition: $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$. Classify the result as a tautology, contradiction, or contingency.

P	Q	$P \rightarrow Q$	$\neg P$	$\neg(P \rightarrow Q)$	$(\neg P \vee Q) \leftrightarrow (\neg(P \rightarrow Q))$
F	F	T	T	T	T
F	T	T	T	T	T
T	F	F	F	F	T
T	T	T	F	T	T

Recall that $P \rightarrow Q$ is defined as $\neg P \vee Q$. Therefore the biconditional $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$ compares a formula with its own definition.

Since $P \rightarrow Q \equiv \neg P \vee Q$ by definition, both sides are identical for all truth value assignments.

All four rows evaluate to TRUE.

$\therefore$ The proposition is a TAUTOLOGY — it is True for all possible truth values of P and Q.

Problem 2 — Logical Equivalence

Prove that $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ (De Morgan's Law) using a truth table.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$	$\equiv?$
F	F	F	T	T	T	T	✓
F	T	F	T	T	F	T	✓
T	F	F	T	F	T	T	✓
T	T	T	F	F	F	F	✓

$\therefore \neg(P \wedge Q) \equiv \neg P \vee \neg Q$ — columns 4 and 7 are identical for all rows. De Morgan's Law is verified.

Section 2: Set Theory

Problem 3 — Set Operations

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. Find: (a) $A \cup B$, (b) $A \cap B$, (c) $A - B$, (d) $B - A$, (e) $A \oplus B$. Verify the inclusion-exclusion principle.

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ (all elements in A or B)

(b) $A \cap B = \{3, 4, 5\}$ (elements in both)

(c) $A - B = \{1, 2\}$ (in A but not B)

(d) $B - A = \{6, 7\}$ (in B but not A)

(e) $A \oplus B = (A - B) \cup (B - A) = \{1, 2, 6, 7\}$ (in exactly one set)

Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B| = 5 + 5 - 3 = 7 \checkmark$

$\therefore A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cap B = \{3, 4, 5\}$, $A \oplus B = \{1, 2, 6, 7\}$. Inclusion-exclusion verified: $7 = 5 + 5 - 3$.

Problem 4 — Power Set

Find the power set $P(A)$ for $A = \{a, b, c\}$. State its cardinality.

For a set with $|A| = n$ elements, $|P(A)| = 2^n$. Here $n = 3$, so $|P(A)| = 2^3 = 8$.

Listing all subsets by cardinality:

$|S| = 0$: $\emptyset$

$|S| = 1$: $\{a\}, \{b\}, \{c\}$

$|S| = 2$: $\{a, b\}, \{a, c\}, \{b, c\}$

$|S| = 3$: $\{a, b, c\}$

$\therefore P(A)$ has 8 subsets. The number of subsets always equals 2^n where $n = |A|$.

Section 3: Relations and Functions

Problem 5 — Properties of Relations

Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)\}$. Determine whether R is reflexive, symmetric, antisymmetric, and transitive. Classify R .

Reflexive: Check $(a,a) \in R$ for all $a \in A$.

$(1,1) \in R \checkmark, (2,2) \in R \checkmark, (3,3) \in R \checkmark \rightarrow R$ is REFLEXIVE.

Symmetric: For every $(a,b) \in R$, check $(b,a) \in R$.

$(1,2) \in R$ and $(2,1) \in R \checkmark; (2,3) \in R$ and $(3,2) \in R \checkmark \rightarrow R$ is SYMMETRIC.

Antisymmetric: $(a,b) \in R$ and $(b,a) \in R$ requires $a = b$.

$(1,2) \in R$ and $(2,1) \in R$ but $1 \neq 2 \rightarrow R$ is NOT ANTISYMMETRIC.

Transitive: For every (a,b) and (b,c) in R , check $(a,c) \in R$.

$(1,2) \in R$ and $(2,3) \in R \rightarrow$ need $(1,3) \in R$. But $(1,3) \notin R \rightarrow R$ is NOT TRANSITIVE.

$\therefore R$ is Reflexive and Symmetric, but NOT Antisymmetric and NOT Transitive. R is not an equivalence relation (fails transitivity) and not a partial order (fails antisymmetry).

Problem 6 — Equivalence Relation

Define a relation R on integers by **aRb if and only if $a \equiv b \pmod{3}$** . Prove R is an equivalence relation and state its equivalence classes.

Reflexive: $a - a = 0 = 0 \times 3$, so $3 \mid (a - a)$. Thus aRa for all integers a . $\checkmark$

Symmetric: If aRb then $3 \mid (a - b)$. Since $a - b = -(b - a)$, we have $3 \mid (b - a)$, so bRa . $\checkmark$

Transitive: If aRb and bRc then $3 \mid (a - b)$ and $3 \mid (b - c)$.

So $a - c = (a - b) + (b - c)$. Since 3 divides both terms, $3 \mid (a - c)$, so aRc . $\checkmark$

Equivalence Classes:

$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$ (multiples of 3)

$[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$ (remainder 1 when divided by 3)

$[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}$ (remainder 2 when divided by 3)

$\therefore R$ is an equivalence relation (reflexive + symmetric + transitive). It partitions $\mathbb{Z}$ into 3 equivalence classes: $[0], [1], [2]$.

Section 4: Proof by Mathematical Induction

Problem 7 — Sum of First n Natural Numbers

Prove by mathematical induction: $1 + 2 + 3 + \dots + n = n(n+1)/2$ for all positive integers n .

Step 1 — Base Case ($n = 1$):

$$\text{LHS} = 1$$

$$\text{RHS} = 1(1+1)/2 = 1$$

LHS = RHS ✓ Base case holds.

Step 2 — Inductive Hypothesis:

Assume the formula holds for some arbitrary $k \geq 1$:

$$1 + 2 + \dots + k = k(k+1)/2$$

Step 3 — Inductive Step (prove for $n = k+1$):

$$1 + 2 + \dots + k + (k+1) = k(k+1)/2 + (k+1) \text{ [by hypothesis]}$$

$$= (k+1)[k/2 + 1]$$

$$= (k+1)(k+2)/2$$

This is exactly the formula with $n = k+1$. ✓

∴ By the Principle of Mathematical Induction, $1+2+\dots+n = n(n+1)/2$ holds for all $n \geq 1$.

Problem 8 — Sum of Geometric Series

Prove by induction: $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all $n \geq 0$.

Base Case ($n = 0$): LHS = 1, RHS = $2^1 - 1 = 1$ ✓

Inductive Hypothesis: Assume $1 + 2 + \dots + 2^k = 2^{k+1} - 1$ for some $k \geq 0$.

Inductive Step ($n = k+1$):

$$(1 + 2 + \dots + 2^k) + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} \text{ [by hypothesis]}$$

$$= 2 \times 2^{k+1} - 1$$

$$= 2^{k+2} - 1 \text{ ✓}$$

∴ By induction, the geometric series sum $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ holds for all $n \geq 0$.

Section 5: Recursion and Recurrence Relations

Problem 9 — Solving a Recurrence Relation

Solve the recurrence relation $T(n) = 2 \cdot T(n-1) + 1$ with $T(1) = 1$. Find a closed-form expression and verify for $n = 1$ to 5.

Iterative unrolling:

$$T(n) = 2 \cdot T(n-1) + 1$$

$$= 2[2 \cdot T(n-2) + 1] + 1 = 4 \cdot T(n-2) + 3$$

$$= 2^2 \cdot T(n-2) + 2^1 + 2^0$$

$$= 2^k \cdot T(n-k) + 2^{k-1} + \dots + 2^0$$

$$\text{Setting } k = n-1: T(n) = 2^{n-1} \cdot T(1) + (2^{n-1} - 1) = 2^{n-1} + 2^{n-1} - 1 = 2^n - 1$$

Verification: $T(1)=1$, $T(2)=3$, $T(3)=7$, $T(4)=15$, $T(5)=31$

All match $2^n - 1$: 1, 3, 7, 15, 31 ✓

∴ Closed-form solution: $T(n) = 2^n - 1$. (This is also the exact number of moves in the Tower of Hanoi problem!)

Problem 10 — Fibonacci Recurrence

The Fibonacci sequence is defined by $F(0) = 0$, $F(1) = 1$, $F(n) = F(n-1) + F(n-2)$. List the first 10 terms and prove that $F(1)+F(2)+\dots+F(n) = F(n+2) - 1$.

n	0	1	2	3	4	5	6	7	8	9
F(n)	0	1	1	2	3	5	8	13	21	34

Proof by induction that $F(1)+F(2)+\dots+F(n) = F(n+2) - 1$:

Base Case (n=1): $F(1) = 1$, $F(3) - 1 = 2 - 1 = 1$ ✓

Inductive Step: Assume $F(1)+\dots+F(k) = F(k+2)-1$.

$$F(1)+\dots+F(k)+F(k+1) = F(k+2)-1+F(k+1) = [F(k+1)+F(k+2)]-1 = F(k+3)-1 \quad \checkmark$$

∴ First 10 Fibonacci terms: 0,1,1,2,3,5,8,13,21,34. Sum identity $F(1)+\dots+F(n) = F(n+2)-1$ proven by induction.

Section 6: Combinatorics and Counting

Problem 11 — Permutations and Combinations

(a) In how many ways can 5 students be arranged in a row? (b) How many 3-member committees can be chosen from 8 students? (c) How many 4-digit PINs can be made from digits 1–9 if digits cannot repeat?

(a) Permutation — order matters, no repetition:

$$P(5,5) = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ arrangements}$$

(b) Combination — order does NOT matter:

$$C(8,3) = 8! / (3! \times 5!) = (8 \times 7 \times 6) / (3 \times 2 \times 1) = 336 / 6 = 56 \text{ committees}$$

(c) Permutation from 9 digits, choosing 4:

$$P(9,4) = 9! / (9-4)! = 9! / 5! = 9 \times 8 \times 7 \times 6 = 3024 \text{ PINs}$$

∴ (a) 120 arrangements (b) 56 committees (c) 3,024 PINs

Problem 12 — Pigeonhole Principle

A drawer contains socks of 4 colors. (a) How many socks must you pick (in the dark) to guarantee a matching pair? (b) How many to guarantee 3 socks of the same color? (c) A class has 367 students — prove at least two share a birthday.

Pigeonhole Principle: If n items are placed in k containers, at least one container has $\lceil n/k \rceil$ items.

(a) Guarantee a pair (2 of same color): 4 colors = 4 pigeonholes.

Worst case: pick one of each color (4 socks). The 5th sock must match one.

Answer: $4 + 1 = 5$ socks (general formula: $k + 1$)

(b) Guarantee 3 of same color: Worst case: 2 of each color = 8 socks.

The 9th sock must give 3 of one color. Answer: **9 socks** (general: $k(m-1) + 1$)

(c) 367 students, 366 possible birthdays (including Feb 29):

By Pigeonhole: 367 pigeons, 366 holes → at least $\lceil 367/366 \rceil = 2$ share a birthday. ✓

∴ (a) 5 socks (b) 9 socks (c) With 367 students and only 366 possible birthdays, at least two must share a birthday by the Pigeonhole Principle.

Section 7: Tower of Hanoi

Problem 13 — Tower of Hanoi: Recurrence, Proof, and Move Sequence

(a) State the recurrence relation for the minimum number of moves $T(n)$ to solve the Tower of Hanoi with n disks. (b) Prove by induction that $T(n) = 2^n - 1$. (c) Write out all moves for $n = 3$ disks (Pegs: $A \rightarrow C$ using B as auxiliary).

(a) Recurrence Relation:

$T(1) = 1$ (base case: one disk, one move)

$T(n) = 2 \cdot T(n-1) + 1$ (move $n-1$ disks to aux, move largest, move $n-1$ disks to target)

(b) Proof by Induction that $T(n) = 2^n - 1$:

Base Case ($n=1$): $T(1) = 1 = 2^1 - 1 = 1$ ✓

Inductive Hypothesis: Assume $T(k) = 2^k - 1$ for some $k \geq 1$.

Inductive Step: $T(k+1) = 2 \cdot T(k) + 1 = 2 \cdot (2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1$ ✓

Step	Disk	From	To	Explanation
1	Disk 1	A	C	Move smallest disk $A \rightarrow C$
2	Disk 2	A	B	Move middle disk $A \rightarrow B$
3	Disk 1	C	B	Move smallest disk $C \rightarrow B$
4	Disk 3	A	C	Move largest disk $A \rightarrow C$
5	Disk 1	B	A	Move smallest disk $B \rightarrow A$
6	Disk 2	B	C	Move middle disk $B \rightarrow C$
7	Disk 1	A	C	Move smallest disk $A \rightarrow C$

$\therefore T(3) = 2^3 - 1 = 7$ moves. Every move is forced — this is the unique optimal solution.